

Paper Circuit LED Greeting Card

Learn How to Light Up Your Creativity!

Subject	STEM / Science & Art
Age Group	Grades 2 – 5 (Ages 7 – 11)
Duration	45 – 60 minutes
Materials	LED, Copper Tape, Coin Cell Battery, Cardstock

Lesson Overview

In this activity, students will explore the basics of electricity and circuits by building a paper circuit that lights up an LED on a handmade greeting card. Using just copper tape, an LED, and a coin cell battery, young learners discover how energy travels through a closed loop — and create a beautiful card to give away!

Learning Objectives	Key Vocabulary
• Define electricity and circuit • Identify circuit components • Build a working paper circuit • Understand LED polarity (+/-) • Apply creativity through decoration	Electricity • Circuit • Battery Copper Tape • LED (Light Emitting Diode) Conductor • Polarity • Positive (+) Negative (-) • Closed Circuit

Science Concepts

What is Electricity?

Electricity is a form of energy caused by the movement of tiny particles called **electrons**. We cannot see electrons, but we can see what they do — they power lights, toys, phones, and almost every gadget around us!

- Powers lights and lamps
- Makes motors spin (toys, fans)
- Runs gadgets and devices (phones, tablets)
- Heats up coils in toasters and kettles

What is a Circuit?

A circuit is a **complete, closed path** that electricity travels along. Think of it like a race track — the electrons zoom around the loop continuously. If there is a gap anywhere, the electrons stop and the light goes out.

Component	Role	In our card
Battery	Gives the power	CR2032 coin cell
Copper Tape	Carries electricity	Strips on card
LED Light	Lights up	Single LED

How Electricity Flows

In our card, electricity travels in a loop:



Materials Needed

#	Item	Quantity	Notes
1	LED Light	1 per student	Any colour; 5mm size works well
2	Copper Tape	~30 cm	Conductive adhesive side down
3	Coin Cell Battery	1 per student	CR2032 (3V)
4	Cardstock / Paper	1 sheet	A5 or A4 folded in half
5	Scissors	1 pair	Adult supervision required
6	Pencil + Ruler	1 set	For planning the circuit path
7	Decorating supplies	As needed	Markers, stickers, crayons

Safety First!

- Always ask an adult for help
- Do not touch batteries for extended periods — they can get warm
- Use scissors carefully with adult supervision
- Keep the circuit dry and away from water
- Do not place coin cell batteries in mouth — choking hazard

Step-by-Step Instructions

1 Prepare Your Paper

- Pick thick cardstock or heavy paper
- Fold it neatly in half — this becomes your card
- Press the fold firmly to create a crisp edge
- Decide where on the card you want your LED to shine

2 Plan Your Circuit Path

- Lightly sketch the copper tape path in pencil first
- The path must form a complete loop from battery to LED and back
- Keep both tape tracks parallel — they must NOT touch each other
- Mark where the LED will sit and where the battery will go

3 Add Copper Tape

- Peel and stick the copper tape along your pencil lines
- Press firmly as you lay it down — good contact is key
- Keep the tape in a straight, connected line
- Do NOT let the positive (+) and negative (–) tracks cross!

4 Place the LED Light

- An LED has two legs — a long one (+) and a short one (–)
- Place the LONG leg on the positive (+) copper tape track
- Place the SHORT leg on the negative (–) copper tape track
- Press the legs firmly against the tape for good contact

5**Add the Coin Cell Battery**

- Slide the battery between the two tape flaps you left
- The SHINY / + side faces UP (touching the positive tape)
- Press the tape flap down firmly over the battery
- The LED should light up — you've completed the circuit!

6**Decorate Your Card!**

- Draw a design that makes the LED part of your picture
- Ideas: lighthouse, star, campfire, sun, firework, fairy light
- Add stickers, colour with markers, or write a message inside
- Share your glowing card with someone special!

Troubleshooting Guide

Problem	Likely Cause	Solution
LED does not light up	Polarity reversed	Flip the LED so the long (+) leg is on the + tape
LED flickers	Poor tape contact	Press tape firmly; smooth out any folds or gaps
LED still off after checking	Broken tape path	Trace the tape — look for gaps or tears; patch with a new strip
Battery gets warm	Short circuit (tracks touching)	Carefully check that the two tape tracks do NOT cross each other
LED very dim	Weak battery or bad contact	Try a fresh CR2032; re-press LED legs onto tape

Teacher Tip

Encourage students who are struggling — troubleshooting IS the learning!
 Real engineers face the same problems. Celebrate 'why didn't it work?' moments.
 A non-working circuit is never a failure — it's an experiment waiting to be solved.

Fun Facts About LEDs and Circuits

- ★ **LED = Light Emitting Diode**
 LEDs are semiconductor devices that emit light when current flows through them. They were invented in 1962 and are now used in virtually every screen and light source.
- ★ **LEDs are energy-efficient**
 LEDs use up to 90% less energy than traditional incandescent bulbs and can last 25,000 hours or more — that is over 10 years of daily use!
- ★ **Circuits are everywhere**
 Every electronic device — phones, game consoles, cars, washing machines — contains circuits. Your phone alone has billions of tiny circuit components!
- ★ **The first electric circuit**
 Alessandro Volta invented the first battery (the Voltaic Pile) in 1800, enabling scientists to create the first controlled electric circuits.

Extension & Creative Project Ideas

➔ Birthday Card with Blinking Lights

Add a resistor and capacitor to make the LED blink on and off — perfect for a birthday or celebration card!

➔ Holiday Decorations

Use multiple LEDs on a larger card or paper panel to create a festive light-up decoration for Diwali, Christmas, Eid, or any occasion.

➔ Storybook with Light-Up Illustrations

Create a short story booklet where the LED lights up a scene — a glowing moon, a lighthouse beam, a campfire, or a twinkling star.

➔ Series vs Parallel Circuits

As an advanced activity, ask students to connect TWO LEDs. Try them in series (one after the other) vs. parallel (side by side) and compare brightness.

➔ Switch Design Challenge

Challenge students to design a simple paper switch using a folded flap of copper tape so the card opens and closes the circuit automatically.

Assessment Checklist

Use this checklist to assess student understanding and project completion:

	Criterion	Notes
■	Student can explain what electricity is in their own words	
■	Student identifies battery, conductor, and LED roles	
■	Student lays copper tape in a continuous, unbroken path	
■	Student places LED with correct polarity (+/-)	
■	Circuit works — LED lights up when battery is inserted	
■	Student can troubleshoot at least one problem independently	
■	Card is decorated with a personal, creative design	
■	Student can explain how electricity flows through the circuit	

Teacher Notes & Tips

Before the lesson

- Pre-cut copper tape into 30 cm strips to save time
- Test each LED and battery pair in advance
- Prepare a completed demo card to show students
- Sketch a simple circuit diagram on the board

During the activity

- Walk around and check LED polarity — this is the most common mistake
- Remind students to press tape firmly at every corner
- Encourage peer learning: a student who succeeds can help a neighbour
- Take photos of student work for a classroom display

After the lesson

- Discuss: 'What was hard? What surprised you?'
- Connect to real-world applications (phone screens, car headlights)
- Send cards home — a great conversation starter with families!
- Store leftover copper tape in a sealed bag to prevent oxidation